

Assignment 2: AI-Based Requirement Extraction from Emails

Problem Statement

In software projects, stakeholders often communicate requirements through emails. These emails may contain incomplete, unstructured, or vague information, making requirement identification difficult and error-prone.

Develop an AI-powered system that connects to Gmail, reads stakeholder emails, analyzes vague or informal conversations, and automatically generates:

- Functional Requirements (FR)
- Non-Functional Requirements (NFR)

The system should intelligently identify software requirements hidden within email conversations and generate structured requirement specifications.

Objective

The objective of this assignment is to:

- Understand AI-assisted Requirement Engineering from textual communication
 - Extract useful information from unstructured email data
 - Use Generative AI/LLMs for requirement analysis
 - Identify and classify Functional and Non-Functional Requirements
 - Handle vague and incomplete stakeholder inputs intelligently
 - Generate structured software requirement outputs
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Features to Implement

1. Connect with Gmail using API integration
2. Fetch and read email conversations
3. Filter relevant project-related emails
4. Clean and preprocess email content
5. Detect vague or incomplete requirements
6. Extract Functional Requirements (FR)
7. Extract Non-Functional Requirements (NFR)
8. Categorize NFRs (Security, Performance, Reliability, etc.)

9. Display generated requirements in structured format
10. Export generated requirements as PDF/DOCX/TXT